Table 1. Descriptive statistics of the four crash subsets, Thailand, 2016–2023.

Variable	S1: Single motorcycle	S2: Motorcycle and another vehicle	S3: Single vehicle, no motorcycle	S4: Multi-vehicle, no motorcycle
Number of crashes (percentage serious injury/fatal)				
2016	1,874 (51.9%)	2,617 (42.5%)	7,482 (15.8%)	3,509 (19.8%)
2017		2,269 (47.6%)	7,911 (13.6%)	3,217 (20.2%)
2018	1,932 (51.2%)	2,759 (47.1%)	8,693 (13.0%)	3,505 (20.7%)
2019		2,583 (52.0%)	8,808 (13.1%)	4,281 (17.8%)
2020	1,791 (52.9%)	3,327 (44.4%)	9,667 (12.3%)	4,933 (15.7%)
2021		2,681 (45.5%)	9,882 (10.2%)	4,189 (15.4%)
2022	1,728 (49.9%)	2,511 (44.1%)	10,320 (9.3%)	4,647 (14.1%)
2023		2,624 (45.7%)	10,968 (9.3%)	5,365 (13.3%)
All years, 2016–2023	**7,325 (51.5%)**	**21,371 (46.0%)**	**73,731 (11.8%)**	**33,646 (16.7%)**
Explanatory variables, mean (standard deviation)				
Vehicle involvement				
One motorcycle involved	–	0.973 (0.163)	–	–
One passenger car involved	–	0.420 (0.494)	0.294 (0.455)	0.361 (0.480)
Two or more passenger cars involved	–	–	–	0.235 (0.424)
One pickup truck involved	–	0.395 (0.489)	–	0.402 (0.490)
Two or more pickup trucks involved	–	0.019 (0.136)	–	0.229 (0.420)
One 6-wheel truck involved	–	0.039 (0.194)	0.049 (0.217)	0.091 (0.288)
One 10-wheel truck involved	–	0.032 (0.177)	0.040 (0.196)	0.083 (0.276)
One truck with more than 10 wheels involved	–	0.074 (0.262)	0.104 (0.305)	0.168 (0.374)
Two or more trucks with more than 10 wheels involved	–	–	–	0.030 (0.171)
Van involved	–	0.023 (0.150)	0.016 (0.126)	0.059 (0.235)
Bus involved	–	–	–	0.025 (0.155)
Pedestrian involved	0.038 (0.192)	–	0.014 (0.119)	–
Crash type				
Rear-end collision in the same lane	0.048 (0.213)	0.323 (0.468)	0.008 (0.091)	0.521 (0.500)
Ran off road to the left and hit an object	0.063 (0.244)	–	0.195 (0.396)	–
Vehicle mounted a traffic island	0.048 (0.214)	–	0.174 (0.379)	–
Ran off road to the right and hit an object	–	–	0.174 (0.379)	–
Sideswipe (parallel movement)	0.033 (0.178)	0.169 (0.375)	–	0.113 (0.316)
Ran off road to the left	0.248 (0.432)	–	0.073 (0.260)	–
Ran off road on a right bend and hit an object	–	–	0.090 (0.286)	–
Ran off road on a left bend and hit an object	–	–	0.073 (0.261)	–
Ran off road to the right	0.067 (0.251)	–	0.047 (0.211)	–
Head-on collision	–	0.054 (0.225)	–	0.073 (0.261)
Loss of vehicle control	0.143 (0.350)	–	–	–
Ran off road on a left bend	0.057 (0.232)	–	0.033 (0.180)	–
Ran off road on a right bend	0.064 (0.246)	–	0.031 (0.174)	–
Crash cause				
Speeding	0.762 (0.426)	0.626 (0.484)	0.774 (0.418)	0.771 (0.420)
Cutting in too closely	0.056 (0.229)	0.276 (0.447)	0.017 (0.128)	0.114 (0.318)
Driver fell asleep	0.021 (0.145)	–	0.104 (0.305)	0.032 (0.177)
Drunk driving	0.103 (0.303)	–	–	–
Running a red light	–	0.029 (0.168)	–	–
Roadway characteristics				
Crash on the main carriageway	0.045 (0.207)	0.052 (0.222)	0.058 (0.234)	0.115 (0.319)
Crash on a frontage road	0.045 (0.208)	0.078 (0.269)	–	–
Road under construction	0.035 (0.184)	0.025 (0.156)	0.018 (0.133)	0.031 (0.173)
Painted median	0.076 (0.266)	0.101 (0.302)	0.051 (0.221)	0.050 (0.219)
Raised or earth median	0.244 (0.430)	0.283 (0.450)	0.255 (0.436)	0.205 (0.404)
Depressed median	0.177 (0.382)	0.179 (0.384)	0.320 (0.467)	0.229 (0.420)
Barrier median	0.101 (0.301)	0.139 (0.346)	0.143 (0.350)	0.341 (0.474)
Horizontal curve	0.176 (0.381)	0.072 (0.258)	0.238 (0.426)	0.082 (0.274)
Concrete pavement surface	0.134 (0.340)	0.155 (0.362)	0.115 (0.319)	0.168 (0.374)
Crest vertical curve	–	–	0.022 (0.147)	–
Sag vertical curve	–	–	0.017 (0.130)	–
On grade	–	–	0.046 (0.210)	0.022 (0.146)
T-intersection	0.028 (0.166)	0.068 (0.251)	0.028 (0.165)	0.046 (0.210)
Four-leg intersection	–	0.054 (0.225)	–	0.050 (0.218)
U-turn without auxiliary lane	–	0.025 (0.155)	–	–
U-turn with deceleration lane only	–	0.027 (0.162)	0.025 (0.155)	0.020 (0.138)
U-turn with deceleration and acceleration lanes	–	0.019 (0.137)	0.020 (0.142)	0.015 (0.122)
Public or commercial access point	–	0.038 (0.192)	–	–
Bridge section	–	–	0.016 (0.125)	–
Environmental characteristics				
Wet road surface	0.057 (0.231)	0.050 (0.218)	0.208 (0.406)	0.109 (0.311)
Raining	0.048 (0.213)	0.043 (0.203)	0.195 (0.396)	0.099 (0.299)
Dark, no street lighting	0.156 (0.363)	0.079 (0.270)	0.082 (0.275)	0.063 (0.242)

Variable	S1: Single motorcycle	S2: Motorcycle and another vehicle	S3: Single vehicle, no motorcycle	S4: Multi-vehicle, no motorcycle
Dark, with street lighting	0.401 (0.490)	0.274 (0.446)	0.387 (0.487)	0.298 (0.457)
Temporal characteristics				
Weekend	0.327 (0.469)	0.280 (0.449)	0.295 (0.456)	0.295 (0.456)
Songkran festival period	0.199 (0.400)	0.103 (0.304)	0.045 (0.207)	0.052 (0.221)
New Year festival period	0.242 (0.428)	0.134 (0.341)	0.049 (0.216)	0.072 (0.259)
Public holiday	–	0.018 (0.133)	0.023 (0.149)	–
Evening	0.347 (0.476)	0.284 (0.451)	0.224 (0.417)	0.236 (0.424)
Midnight / early morning	0.240 (0.427)	0.094 (0.291)	0.288 (0.453)	0.157 (0.363)
Peak hours	0.173 (0.378)	0.255 (0.436)	0.175 (0.380)	0.199 (0.399)
Region				
Eastern region	0.068 (0.251)	0.065 (0.247)	0.123 (0.328)	0.154 (0.361)
Northeastern region	0.197 (0.398)	0.188 (0.391)	0.223 (0.416)	0.147 (0.354)
Northern region	0.168 (0.373)	0.127 (0.333)	0.128 (0.334)	0.072 (0.258)
Southern region	0.134 (0.341)	0.137 (0.344)	0.166 (0.373)	0.102 (0.303)
Western region	0.070 (0.256)	0.064 (0.245)	0.085 (0.278)	0.060 (0.237)

Notes. Outcome: serious injury/fatal (1) vs. property damage only/minor injury (0). S1 is analysed in two-year periods (2016–2017, 2018–2019, 2020–2021, 2022–2023) because of its smaller sample; S2–S4 in yearly periods. All explanatory variables are indicators (1 if yes, 0 otherwise), so the mean is the share of crashes with the indicator; standard deviation is the sample standard deviation. – : not a candidate variable in that subset. Reference categories: Central region, daytime, no median (and one motorcycle in S1, one pickup truck in S3).